NFL Coaching Value

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Abstract

While coaching quality is a primary driver of success in the NFL, a standard quantitative metric to measure a coach's literal value to their team remains elusive. This research introduces a data-driven framework to quantify coaching value by analyzing play-by-play data since 2017. Coaches are evaluated through a cumulative score derived from several independent performance categories, including decision-making in high-leverage situations (fourth downs, two-minute drills) and raw success (1-score games, regular season, divisional, and playoff records).

A key element of this paper is the isolation of player development using a Hierarchical Bayesian Adjusted Plus-Minus (APM) model based on Expected Points Added (EPA). By comparing observed player impact against expectations based on age, experience, volume, and draft position, systemic roster overperformance is identified and attributed to coaching quality. The framework also offers practical utility by identifying high-performing current coaches like Mike Vrabel and targeting valuable assistant candidates for future hiring cycles.

Furthermore, the model highlights discrepancies between coaching value and job security, as seen in the firing of Sean McDermott despite a strong score in 2025 and the second highest total score since 2017. Ultimately, this research provides NFL organizations with a robust tool to move beyond traditional narratives and objectively identify elite coaching talent.

Introduction

After the 2025 NFL Season, 10 teams were looking for a new Head Coach. The Bills, Cardinals, Falcons, Ravens, Browns, Raiders, Dolphins, Giants, Steelers, and Titans all were looking for the best option to lead their team in a new direction, which begs the question: how do you figure out who the best coach available is? Typically, we would consider the "hottest" up-and-coming assistant coaches who may be an Offensive or Defensive Coordinator for a team that just had a great season. We may consider former head coaches, preferring those who have had very tangible success in the form of Playoff appearances and Super Bowls. Almost every time, you'll hear someone say the candidate came from someone else's "coaching tree," meaning they worked under another prominent coach and therefore maybe some of that success will have rubbed off on this particular candidate. None of these conversations are particularly data-driven, however, and most dialogue about NFL teams' successes do not give coaches the appreciation they deserve.

More so than any other professional sport, NFL coaches are incredibly involved in the play-by-play flow of the game: calling every play, managing personnel, sequencing plays cleverly to trick the defense, etc. The battle between play callers is an entirely hidden part of the game. It's easy enough to try to quantify the value that a player adds in sports. You can consider them as a certain number of points, or a certain number of wins as is done in most MLB statistics. Coaches are trickier. They aren't actively on the field, and there is not necessarily any tangible way to quantify their contributions. As such, there really is no existing metric for how much value NFL Coaches add or any way to directly compare them from a statistical standpoint other than looking at numbers such as wins, playoff appearances, Super Bowls, and Coach of the Year awards. It's also worth considering how well they perform in two-minute drills, on fourth downs, or when they're in a one-score game. How well a coach can maximize the talent they have is what sets them apart, and is critical to a successful season.

This research will attempt to quantify the value that coaches add in various metrics such as performance in critical game states, raw success, and player development. It will then directly compare coaches within each individual season to "grade" them on their performance, resulting in a tangible score representing a coach's success relative to the pack.

Data & Methodology

Data

The data for this research is collected from various sources. Play-by-play data is used from the R packages nflfastR and nflreadr. These libraries include relevant information such as play results, EPA and WPA on every play, participating players on plays, and additional play details. Additionally, these libraries have information on team's season records as well as playoff performances. Data for team's coaches (head coach, offensive coordinator, and defensive coordinator) were collected from PFR. These were collected by team-season, where a row represents the three aforementioned positions for a given team in a given season.

Methodology

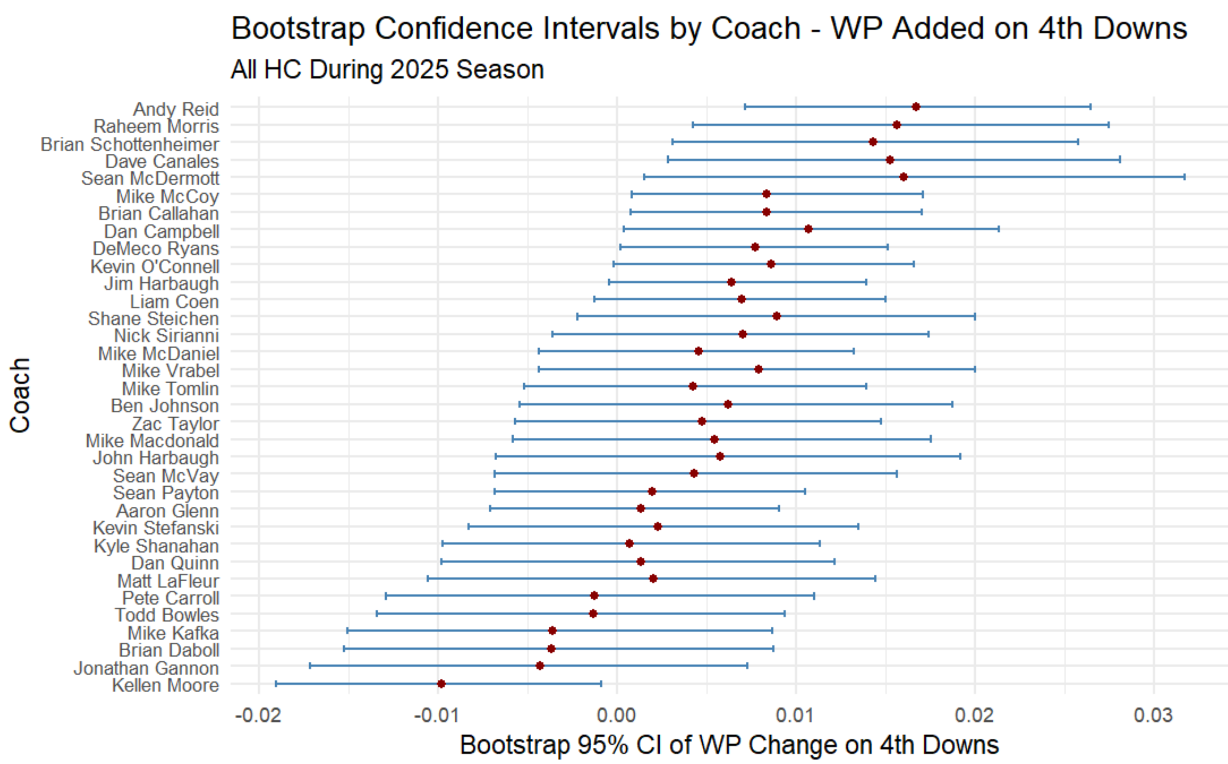
Overall Approach

To evaluate coaching performances, we can aggregate a series of various metrics measuring the value coaches add in different ways. The metrics I will be utilizing in this research are success in certain game types, value added in certain game states, and ability to maximize talent. These "scores", when combined, result in one larger score that reflects a coach's all around ability as well as the success of the team he leads.

Fourth Down

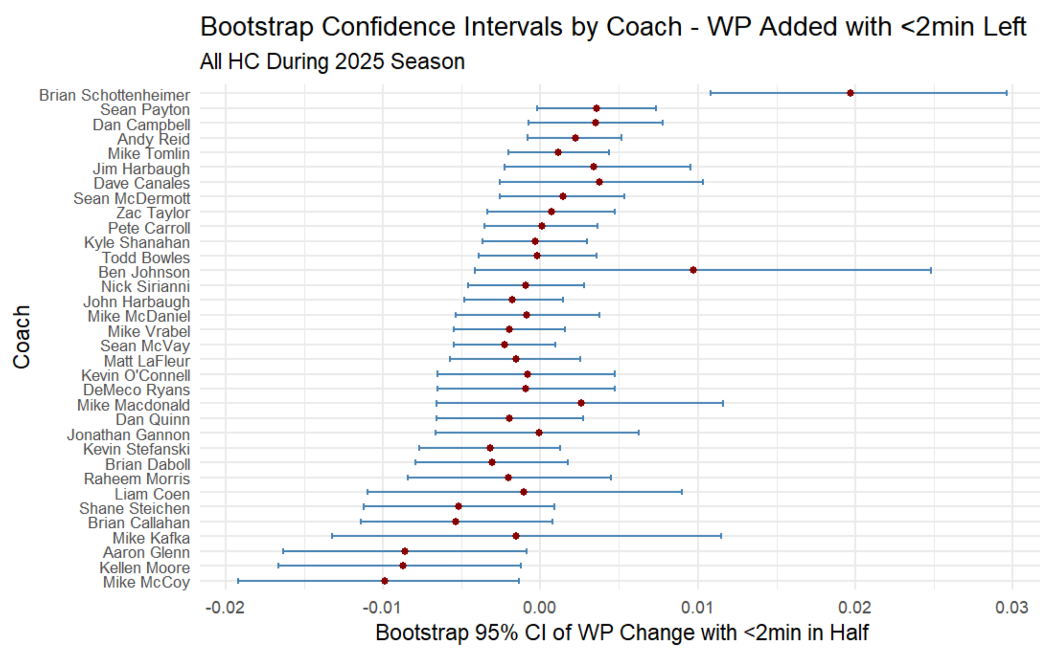
Starting off with game states, we have fourth downs: the most frequent situation where a coach must make a big decision. Typically, these decisions boil down either to punts, field goals, or “going for it” and trying to convert to get a new set of downs. Being able to make the most of a bad situation is what can make or break a drive and even a game. To measure a coach’s "aggressiveness score”, or really just their ability to make the most of a fourth down situation, we will look at win probability added in such game states, and then bootstrap the results to create intervals that represent where a coach’s true mean lies for this metric. An interval that does not overlap zero indicates a likely true ability to either add or subtract value in these situations, and also allows us to directly compare coaches.

On fourth downs in the 2025 season, head coaches Andy Reid, Raheem Morris, Brian Schottenheimer, and Dave Canales led on this metric, each with an interval entirely above 0, indicating that they all add value on fourth downs. Only one coach had an interval entirely *below* zero: Kellen Moore. This metric is skewed positively, so most coaches have a mean above 0. They were scored based on their percentile, where the highest mean wp% added would be a score of 1, and the lowest a score of 0. In this case, Andy Reid would be the highest scoring coach this season on this metric, and Moore would be the lowest.



Crunch Time

Arguably the part of the game most reliant on good coaching is in the final two minutes of a half. Essentially, coaches want to get the last word in without allowing the other team a chance to respond. To measure this, we will use the same method as for fourth downs. In 2025, Brian Schottenheimer, rookie coach for the Dallas Cowboys, led the pack by a wide margin, with an interval ranging from .011 (1.1% wp added per play) to .03 (3% wp added per play). In second was Sean Payton (.003, .026). Pete Carroll (-.023, .08) and John Harbaugh (-.025, .002) trail the pack, and both were let go this offseason.



Coaches were scored based on their percentile mean wp% added, with coaches with the highest wp% added receiving a score of 1, and those with the lowest receiving a score of 0. These were once again grouped by season, so coaches were compared to the field for that given season. This season, Schottenheimer would have been the highest scoring, and Harbaugh would have been the lowest.

Close Games

The ability to win close games is often what makes or breaks a team in a given year. In 2024, the Kansas City Chiefs were 11-0 in one score games, tied with the 2022 Vikings for the best record in such games. They would go on to win the AFC before losing in the Super Bowl. The following season, in a much more disappointing year, they were the worst team in one score games, going only 1-9. All of the teams who fired a head coach this past season went a collective 29-59 (.329) in such games. The only teams with a winning record in one-score games to fire their head coach were the Buffalo Bills (5-3) and the Miami Dolphins (4-3).

Since 2015, playoff teams are 807-457 (.638), with 122 out of 145 (84.1%) teams making the playoffs having a .500 or better record in such games. Teams who make the Super Bowl are 125-54 (.698) with only 2 teams without a winning record in such games having made one. Both of those teams lost.

That is to say, winning close games matters. Coaches were scored based on their percentile ranking for that given season, with the coaches with the highest win percentage in close games in a given season receiving a score of 1, and those with the lowest receiving a score of 0. Top performing active coaches with at least 3 seasons coached since 2015 are Mike Tomlin, DeMeco Ryans, and Nick Sirianni, who all place, on average, around the 75th percentile in their seasons. Bottom performing coaches with the same criteria are Raheem Morris, Brian Daboll, and Jonathan Gannon, each of whom were fired this past offseason.

Division Games

The most important games of the regular season tend to be Divisional games. These are the teams that you see twice a year, and are typically a crapshoot as far as predicting who will win goes. All bets are off, and it tends to come down to the margins and preparation before the game. No team since 2015 has made a Super Bowl without a .500 or better divisional record. Only 7 teams have even made the playoffs without a .500 or better divisional record. These are, of course, inherent to the nature of making the playoffs (winning games is good), but is still worth noting. Teams who fired their head coaches this season were a collective 20-46 in divisional games. Three such coaches had a .500 record (Harbaugh, McDaniel, Morris), and Sean McDermott was 4-2 in such games.

Coaches were scored based on their percentile win percentage in division games, with coaches with the highest win percentage in divisional games receiving a score of 1, and those with the lowest receiving a score of 0.

Playoffs

Arguably the most important thing a coach can do is get his team to the playoffs and give them a chance to play for a Super Bowl. So, the next metric is focused on deriving playoff success from playoff performance, and scoring coaches on their ability to both get *to* and *through* the playoffs. By assigning each stage of progression as a "tier" where winning the Super Bowl is tier 6, and each subsequent level is one tier lower, with missing the playoffs entirely being a 1. Then, scoring them by taking the percentile of the tier that they reached.

Only one coach that made the postseason this year was fired (Sean McDermott), and historically, getting to the playoffs typically does save you your job even if you're on the hot seat. The other 10 coaches fired were all among the 18 teams on the outside looking in this postseason. Since 2017, the best head coaches (n >3 seasons) on this metric are Andy Reid, DeMeco Ryans, and

Nick Sirianni. The worst are the four coaches who have failed to make the playoffs in that span: Jonathan Gannon, Mike McCoy, Raheem Morris, and Shane Steichen.

Player Development & Talent Maximization

The final metric for this score is a coach's ability to get their players to overperform expectations based on past performance and workloads.

Because football is such a highly interactive sport in which every play involves 22 players simultaneously influencing the outcome, evaluating the individual contribution of a single player is difficult. To address this issue, I used a Hierarchical Bayesian Adjusted Plus–Minus (APM) model to estimate the effect of every player on the field during each play.

In this framework, each NFL play is treated as an observation, and the outcome of the play is measured using Expected Points Added (EPA). EPA estimates how much a play increases or decreases a team's expected scoring potential given the game situation. Positive EPA indicates a successful play for the offense, while negative EPA indicates success for the defense. The goal of the model is to explain the EPA of each play as the combined influence of all players on the field, while accounting for random variation in play outcomes.

Let N denote the number of plays in the dataset (approximately 40-50,000 per year), P denote the number of players who appear in those plays, and y_i denote the EPA outcome of play i. Let X be the matrix of vectors indicating which players participated in a given play where X_{ij} is 1 or -1 depending on whether player j is on offense or defense during play i, or 0 otherwise. Each row of X then represents the set of players participating in a given play, while each column corresponds to a single player.

The outcome of a play is modeled as a linear combination of player effects:

$$y_i \sim Normal(\alpha + \sum_{j=1}^{P} X_{i,j}\beta_j, \sigma)$$

where y_i is the EPA of play i, α is the baseline EPA of a play, β represents the contributions of player j, and σ represents random variation in play outcomes. Intuitively, this model states that the result of a play is determined by the combined contributions of the players plus involved random noise. Positive player effects increase offensive success, while negative effects indicate that the player tends to reduce offensive success.

A potential challenge when estimating player effects is that many NFL players are backups that see very few relative snaps. Without safeguards, the model could produce extreme estimates for them. To address this, the model uses hierarchical Bayesian regularization, which assumes that individual player effects are drawn from a common distribution centered near zero:

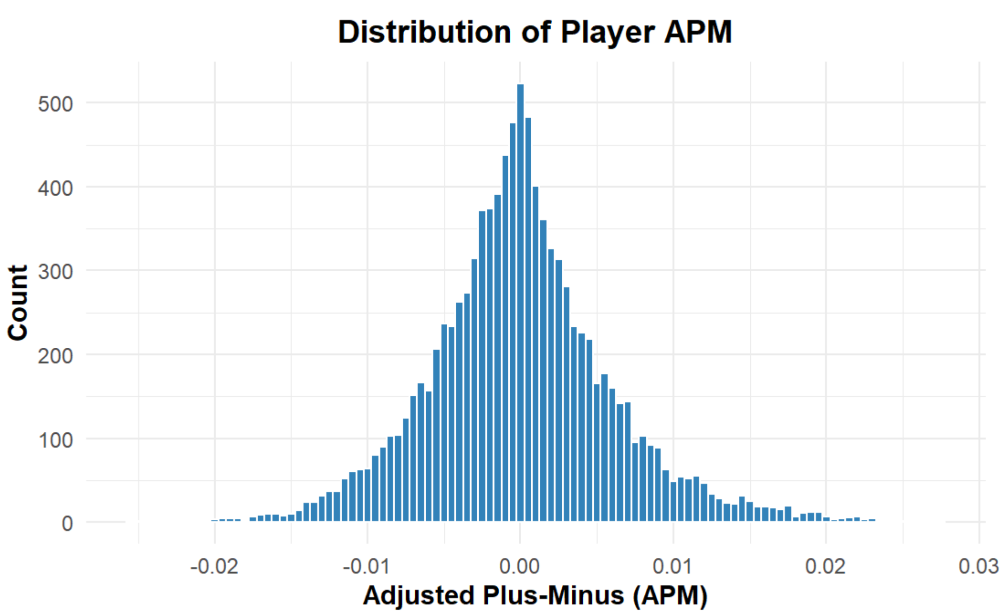
$$\beta_j \sim Normal(0,\ \tau)$$

where τ represents the variability of player talent across the league. This assumption reflects the belief that most players are relatively close to league average, while allowing exceptional players to stray from that average when supported by enough data. This structure produces shrinkage, where estimates for players with limited data are pulled towards the mean, and estimates for players with a large number of snaps remain primarily determined by the observed data.

This Bayesian framework requires specifying prior distributions for model parameters:

$$\alpha \sim\ Normal(0,\ 10)$$
$$\tau \sim\ Normal(0,\ 10)$$
$$\sigma \sim\ Normal(0,\ 10)$$

These priors are weakly informative and reflect reasonable assumptions about the magnitude of play outcomes and player effects. The model parameters are estimated using Markov Chain Monte Carlo (MCMC) simulation, where we repeatedly sample from the posterior distribution of the parameters given the observed data. Rather than producing a single point estimate, we instead generate a distribution of plausible values, and from those we can compute the posterior mean and standard deviation which serves as the player's APM estimate and uncertainty of said estimate. Players who have a lot of snaps have more precise estimates, and those with fewer snaps have wider uncertainty levels. The resulting APM value for player j, denoted as β_j represents the average effect that player has on expected points during a play, controlling for all other players on the field. Below is the distribution of Player APMs since 2016.



In order to apply these APM estimates to coaching, I then modeled expected player performance and measured deviations from that expectation. This approach makes the assumption that if players systematically outperform these expectations, that this may indicate a positive coaching effect. This performance is modeled using the previously described APM estimates, so to estimate performance in season *t*, I fit a regression model predicting player APM using prior performance, experience, snaps, position, and draft position (primarily meant for young players). The model is specified as:

$$APM_{i,t} = \beta_0 + \beta_1 APM_{i,t-1} + \beta_2 Exp_{i,t-1} + \beta_3 Exp^2_{i,t-1} + \beta_4 DraftPos_i$$
$$+ \beta_5 log(Snaps_{i,t-1} + 1) + \beta_6 log(Snaps_{i,t} + 1) + \beta_7 Exp_{i,t} + \gamma Position_i + \epsilon_{i,t}$$

where APM represents last year's APM, Exp represents a player's experience, DraftPos represents a player's draft position, Snaps represents last year's and this year's snap counts, and Position represents the player's position. This model captures the typical trajectory of player performance based on observable characteristics.

Predicted APM values from the model represent the expected performance level for each player-season: $\widehat{APM}_{i,t}$ and player development is then defined as the deviation between observed and expected performance: $D_{j,t} = APM_{i,t} - \widehat{APM}_{i,t}$ where positive values indicate players outperforming expectations,while negative values indicate underperformance. These scores are then aggregated across players for each team-season, and separate scores are kept for offensive and defensive players. To emphasize the importance of this metric, the "score" is then kept as three separate scores, each worth one point. There is a total score, offensive score, and defensive score. The percentile ranking is then taken and applied on a scale from 0 to 1. So, a coach with a score of 1 for total development score likely also has a high offensive and defensive score. This allows us to specifically look at one side of the ball, since most coaches are considered either an offense-oriented or defense-oriented coach.

Some of the top seasons since 2017 include the 2019 Ravens (+.018), 2024 Bills (+.016), 2025 Patriots (+.014), and 2024 Eagles (+.014). This season had three of the four worst seasons on this metric, with the Jets (-.015) and Raiders (-.013) posting the two lowest scores since 2017, and the Titans at fourth lowest (-.012), with the Raiders and Titans firing their head coach this offseason.

Results and Discussion

With all of these metrics in place, we can construct a score out of 9 (scaled to 10 points) that allows us to “grade” a coach on a wide variety of criteria. These criteria cover their game management abilities, regular and postseason success, as well as their ability to bring out the most in the talent they are given.

Best Coaches Since 2017

Looking at the past 9 seasons, there’s not a lot of surprises that appear immediately. There’s inherent irony in Sean McDermott being the second highest scoring coach in that span, along with Mike Tomlin and Pete Carroll who both also did not stay with their team after this season. Andy Reid comes in as the top coach, with 2025 being his worst season.

Coach	seasons	Teams	2017	2018	2019	2020	2021	2022	2023	2024	2025	Total
Andy Reid	9		6.02	7.72	8.51	8.71	7.74	7.98	7.57	8.55	3.83	**66.63**
Sean McDermott	9		5.36	2.75	5.56	8.11	6.88	7.62	6.81	8.66	7.89	**59.65**
Sean McVay	9		7.41	8.85	3.79	5.71	7.49	3.08	5.97	5.49	7.24	**55.02**
Sean Payton	8		7.50	8.88	7.54	8.30	4.55		4.11	5.69	7.44	**54.00**
John Harbaugh	9		4.68	5.69	8.87	6.82	3.83	6.05	7.09	6.62	3.46	**53.10**
Mike Tomlin	9		7.86	6.30	4.31	6.48	5.92	5.21	5.35	5.37	5.66	**52.46**
Kyle Shanahan	9		3.33	1.41	9.08	3.45	6.71	7.78	7.38	2.87	6.33	**48.34**
Matt LaFleur	7				8.34	7.86	7.04	3.43	5.11	5.96	5.14	**42.87**
Pete Carroll	8		4.76	6.76	6.56	7.31	3.45	5.48	4.10		1.08	**39.51**
Mike McCarthy	7		4.14	3.77		3.52	7.77	7.91	7.73	3.55		**38.39**

Best Coaches Since 2023

Looking at a more recent timeframe, we can identify the coaches who are doing the best *right now*. Sean McDermott leads the pack, followed closely by relatively newer Head Coach Dan Campbell. DeMeco Ryans also stands out as a newer coach off to some early success.

Coach	seasons	Teams	2023	2024	2025	Total
Sean McDermott	3		6.81	8.66	7.89	**23.36**
Dan Campbell	3		7.92	9.04	5.22	**22.18**
Nick Sirianni	3		6.31	8.70	5.39	**20.40**
Andy Reid	3		7.57	8.55	3.83	**19.95**
DeMeco Ryans	3		6.27	5.60	7.56	**19.43**
Sean McVay	3		5.97	5.49	7.24	**18.69**
Sean Payton	3		4.11	5.69	7.44	**17.24**
John Harbaugh	3		7.09	6.62	3.46	**17.17**
Todd Bowles	3		6.79	6.20	4.01	**17.00**
Kyle Shanahan	3		7.38	2.87	6.33	**16.58**

Best Coaches This Season

Looking even narrower, we can identify the top coaches from this past NFL season. Mike Vrabel comes in with a strong 8.61 in his first year as the HC of the New England Patriots. Ben Johnson had a good showing in his first season as the Chicago Bears' Head Coach, and all other coaches in the top 10 this season were playoff contenders. Only Brian Schottenheimer (DAL, 5.34) and Dan Campbell (DET, 5.22) outscored any playoff contending coaches. Matt LaFleur (GB, 5.14) came in as the lowest scoring playoff coach.

Coach	seasons	Teams	2025	Total
Mike Vrabel	1		8.61	**8.61**
Sean McDermott	1		7.89	**7.89**
Mike Macdonald	1		7.60	**7.60**
DeMeco Ryans	1		7.56	**7.56**
Sean Payton	1		7.44	**7.44**
Sean McVay	1		7.24	**7.24**
Liam Coen	1		7.08	**7.08**
Jim Harbaugh	1		6.66	**6.66**
Ben Johnson	1		6.62	**6.62**
Kyle Shanahan	1		6.33	**6.33**

Worst Coaches This Season

Moving to the bottom of the pack, we have our bottom 10 performers from this season. 6 of the 10 coaches in this list were fired this offseason (MIA, CLE, TEN, NYG, ARI, LV). Aaron Glenn was the worst performer this season with a score of only 0.88.

Coach	seasons	Teams	2025	Total
Zac Taylor	1		3.29	**3.29**
Mike McDaniel	1		3.26	**3.26**
Kevin Stefanski	1		2.78	**2.78**
Kellen Moore	1		2.40	**2.40**
Brian Callahan	1		2.09	**2.09**
Brian Daboll	1		2.01	**2.01**
Dan Quinn	1		1.88	**1.88**
Jonathan Gannon	1		1.39	**1.39**
Pete Carroll	1		1.08	**1.08**
Aaron Glenn	1		0.88	**0.88**

Top Available Non-HC

This framework also allows us to directly look at potential candidates that haven't had any head coach experience. Joe Brady leads the pack, and was hired as the new Bills HC this offseason. Mike LaFleur was hired to the Cardinals and Todd Monken to the Browns. Three other assistant coaches were hired: Jesse Minter (3.33 in 2025) to the Ravens, Klint Kubiak (3.80 in 2025) to the Raiders, and Jeff Hafley (2.57 in 2025) to the Dolphins.

Coach	seasons	Teams	2023	2024	2025	Total
Joe Brady	3		3.40	4.33	3.94	**11.68**
Vic Fangio	3		3.12	4.35	2.69	**10.16**
Matt Nagy	3		3.79	4.27	1.92	**9.98**
Steve Spagnuolo	3		3.79	4.27	1.92	**9.98**
Matt Burke	3		3.14	2.80	3.78	**9.72**
Mike LaFleur	3		2.98	2.74	3.62	**9.35**
Joe Lombardi	3		2.05	2.84	3.72	**8.62**
Vance Joseph	3		2.05	2.84	3.72	**8.62**
Todd Monken	3		3.54	3.31	1.73	**8.59**
Aaron Glenn	2		3.96	4.52		**8.48**

Looking at Specific Metrics

Top 10 Head Coaches Since 2023; Offensive Player Development

Coach	seasons	Teams	2023	2024	2025	Average Score
Sean McDermott	3		9.41	10.00	9.70	**9.70**
Matt LaFleur	3		9.12	7.35	9.09	**8.52**
Brian Schottenheimer	1				8.48	**8.48**
Dan Campbell	3		8.53	9.12	6.97	**8.21**
Sean McVay	3		8.24	5.88	10.00	**8.04**
Kyle Shanahan	3		10.00	6.18	7.88	**8.02**
Mike Vrabel	2		6.18		8.79	**7.48**
John Harbaugh	3		7.94	9.71	4.55	**7.40**
Andy Reid	3		6.47	7.94	7.58	**7.33**
Ben Johnson	1				7.27	**7.27**

Top 10 Head Coaches Since 2023; Defensive Player Development

Coach	seasons	Teams	2023	2024	2025	Average Score
Kevin O'Connell	3		7.35	9.71	9.09	**8.72**
Mike Macdonald	2			7.94	9.39	**8.67**
Josh McDaniels	1		8.53			**8.53**
Liam Coen	1				8.48	**8.48**
Kevin Stefanski	3		10.00	6.47	8.79	**8.42**
Jim Harbaugh	2			8.24	7.58	**7.91**
DeMeco Ryans	3		3.82	8.82	10.00	**7.55**
Bill Belichick	1		7.06			**7.06**
Matt Eberflus	2		8.24	5.88		**7.06**
Kellen Moore	1				6.97	**6.97**

Top 10 Head Coaches Since 2023; Value Added in Critical Game States.

Fourth Downs, 2-Minute Drills.

Coach	seasons	Teams	2023	2024	2025	Average Score
Brian Schottenheimer	1				9.10	**9.10**
Dan Campbell	3		8.57	8.48	7.83	**8.29**
Ben Johnson	1				7.60	**7.60**
Sean McDermott	3		5.19	8.14	8.76	**7.36**
Andy Reid	3		6.70	9.37	5.48	**7.19**
Dave Canales	2			6.22	7.93	**7.08**
Raheem Morris	2			5.04	8.57	**6.80**
Mike McCarthy	2		5.98	6.44		**6.21**
Matt Eberflus	2		5.36	7.02		**6.19**
Todd Bowles	3		8.76	5.68	4.05	**6.16**

Conclusion

Future Work

When initially beginning this research, I had also hoped to include a score for a coach's "network", where coaches are rewarded for their association to other good coaches. In the NFL, "coaching trees" are a very popular conversation point and very well may play a role in teams' hiring decisions. As this project evolved, many of the other metrics used were narrowed in on a single season, and this score would not be able to do that. It would effectively be an "intercept" for the coach, where they are always given some sort of bump for their name and the names of coaches they are associated with. It goes beyond the season scope, so I chose to omit it from the final deliverable. It wouldn't add much information, and goes beyond the scope of this project. Further research should consider the relationships between coaches and how association to a "good" coach may predict future performance and success as a head coach. For the sake of this report, assistant coaches just get half credit for the score of the head coach.

Limitations

The primary limitation of this research was a lack of existing framework for evaluating coaches this way, and a lack of existing datasets for coaching trees. Only primary coaches were readily available, and as such most other coaching positions were not able to be added. This includes anything ranging from Passing Game Coordinator to Linebackers Coach. Only Head Coaches, OC, and DC were included, narrowing the overall scope of the project.

Wrapping Things Up

This framework provides an accessible way to quickly and effectively evaluate a coach based on any of 8 metrics, with custom weights for each one. Using the Companion Shiny Web App allows users to define their own criteria for which of these metrics are more important than others, look at specific time frames, filter by coach types, and then download the results. With this, we can identify top performing head and assistant coaches, ultimately aiding in not only describing how good a coach is, but potentially projecting future performance when making a hiring (or firing) decision. This approach is by no means perfect, but is a good start in adding more nuance into how we evaluate coaches in the NFL, and quantifies the value that they add.